

Week 8 Essay Task

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PSY-IS-4158 ISM- Validation for Substance Abuse Scale for Indian Population

Professor Naseer Ahmad Bhat

Cultural, Linguistic, and Contextual Considerations for Adapting the DAST 10 in India

While the disorder itself may manifest similarly across cultures, how individuals respond to screening questions can vary significantly based on societal perceptions of substance use.

In societies where drug use is heavily stigmatized, individuals may be less likely to respond honestly due to fear of judgment or other consequences (Yang et al., 2017). Additionally, if gender inequalities are prevalent and substance abuse is further stigmatized for women, especially in conservative areas, underreporting may become an even bigger challenge. Conversely, in cultures where recreational drug use is more accepted or even legal, responses may reflect different attitudes toward what constitutes problematic behavior. For example, children in certain cultures may be more strictly monitored by their parents, increasing the possibility of them complaining about their children's involvement with drugs, which is an item in the DAST 10. Similarly, in uneducated and poorer populations, people may be more accepting of substance due to its widespread use (Giri 2024).

In India specifically, translating terms like "substance" requires careful attention. Many individuals may only associate the term with hard drugs like cocaine, rather than traditional or more accessible substances like bhang or unprescribed pharmaceutical drugs (Karki & Rangaswamy 2023). Given the ease of obtaining these medications in India, it's crucial to clarify that misuse of such drugs also constitutes substance abuse (Marathe et al., 2020). Furthermore, the lower levels of legal literacy could lead to underreporting, as people might not fully understand whether their drug procurement was illegal, which is another item in the DAST 10. Socioeconomic disparities pose another significant challenge. Pilot studies often involve more

affluent, educated populations, but when the scale is used among economically disadvantaged groups, it may fail to account for their unique contexts. In these cases, concepts like substance abuse might be unfamiliar, leading to inaccurate or incomplete responses.

To tackle these issues, we may give briefings before the participants take the test which may be in the form of video or audio recordings. The aim here is to minimize interpretational differences. This way, we can ensure that the participants have a good understanding of what “substance abuse” entails and what the various items on the scale actually mean. Furthermore, we can promote digital tests to minimize the effects of stigma. Anonymity will help these individuals to be more honest about their responses and less concerned about facing judgment.

Ethical Barriers and Considerations

When dealing with a substance use disorder patient group, it is expected to face significant ethical challenges which we must take into consideration and account for.

Patient Vulnerability:

Given that these participants are already struggling/ have struggled with substance abuse, it is possible that the interview process may end up reminding them of past traumatic events or cause them to relapse. In order to tackle this, I think it is important to ensure that these individuals are held under close supervision in the psychiatric ward at Institute of Human Behavior and Allied Sciences (IHBAS) to prevent relapse or to provide counseling if necessary. Given that these participants are patients in and are availing services of a particular institution, they may feel forced to participate. Even though informed consent is present, they may feel an underlying pressure to volunteer. It is important to repeatedly reiterate their rights to withdraw and to emphasize that the study is separate from their treatment.

Potential Risks:

There is a risk of making these patients feel judged/ stigmatized as a result of the phrasing of the questions or by the perceived reactions of the interviewer. "Because substance use disorders carry such a stigma, primary care clinicians who screen their patients for substance abuse unavoidably intrude on their autonomy and their privacy. Whether clinicians screen through laboratory testing or by administering behavioral questionnaires, they are seeking very personal information... A patient "in denial" may not realize, or want to realize, that he has to cut

back on or give up his alcohol or other drug use, and he may view the clinician's questions and suggestions as intrusive” (Sullivan & Fleming, 2005). In which case, they may feel uncomfortable or may end up being dishonest. To prevent this, the attending professionals at IHBAS can be the administrators of the tests/ interview post adequate training.

Additionally, if the participants are unaware of the aim of the interview/study, they may personalize its purpose which may lead to other cognitive distortions. We can make the debriefing summary mandatory for all the participants in order to prevent this.

Other suggestions:

The demographic section of the interview can be prior to the assessment section as it'll make it easier for the participants to withdraw if they are uncomfortable sharing personal information. If they have already spent significant time on the interview, they might not feel free to withdraw at the final stages where they disclose demographic information.

Apart from the option to withdraw, we may also allow the participants to skip certain questions that they wish not to answer. If the data points to a particular item or a set of items being frequently skipped by the participants, we may identify shortcomings with regards to the phrasing or any other aspect of that particular item.

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Week 9 Essay Task - 3 Areas for Improvement

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Introduction - Providing the Benefits of Adapting a Pre-existing Scale over Creating a New One and Cultural Differences that Motivate the Adapting of DAST-10

The introduction gives a detailed explanation on the widespread prevalence of substances in India and as well as the lack of English speaking populations. However, there are some unanswered questions - why should we adapt the DAST-10 instead of newly creating an equally comprehensive scale using the others mentioned in the study (ASSIST, ASSORT, etc) and specifically tailor it to the Indian population? But also, why is it necessary to go through an in depth process of adaptation over simple translation? I believe the value of the study will be better illustrated if these questions are answered.

Borsa et al., (2012) mention the benefits of using a pre-existing validated scale in their paper:

By adapting an instrument, the researcher is able to compare data from different samples and from different backgrounds, which enables greater fairness in the evaluation because the same instrument assesses the construct based on the same theoretical and methodological perspectives. The use of adapted instruments naturally enables a greater ability to generalize and also enables one to investigate differences within an increasingly diverse population.

Apart from improving the quality of evaluation for the adapted version, using different versions of the same scale enables us to make cross-cultural studies and compare various groups. This is especially beneficial in a country like India which houses a linguistically diverse population. Eventually, we will be able to make better inferences about the underlying psychological factors (in this case, wrt substance abuse disorder) from a larger group and also

assess how cultural differences influence our substance use patterns through comparisons by using the same template scale. Additionally, we also know that the DAST-10 has already been translated into many other languages and is being used in other countries which makes it more suitable for these kinds of comparative studies.

The need for a thorough adaptation comes from the fact that people carry different biases based on their cultural context. A few examples are as follows:

Construct Bias. The concept or psychological construct being measured may not be equally meaningful across different cultural groups. What is considered a harmful substance or at what point does usage become abuse may change based on one's prior beliefs.

Item Bias. refers to issues at the individual item level within a test. Sociocultural factors affect how individuals understand and respond to particular items. For example, familial neglect may look different to different people based on the values and responsibilities they associate with it (Byrne, 2015).

There are many other factors/ biases that may affect the validity of the scale which is why it is important to thoroughly compare our scale with the pre existing, validated scales and ensure that the scores on the test indicate the same level of severity as intended.

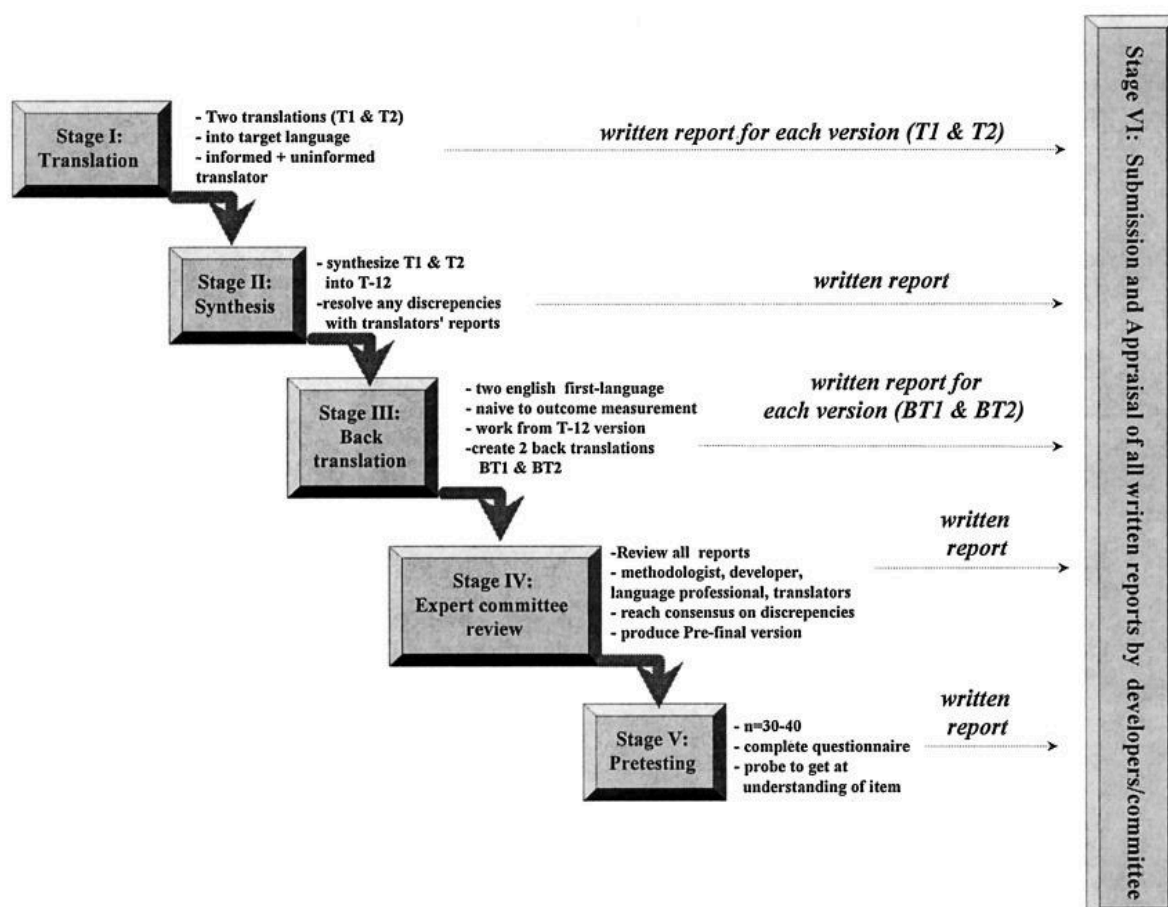
Methods (Design): Elaboration on the Translation/ Adaptation Process

While the method of testing is well outlined in the research proposal, I think the process of adaptation needs to be more elaborate. There can be multiple stages to the process that starts with the translation task already mentioned in the research proposal. Even though it is indicated that there will be a thorough refinement of this translation post this initial step, it is a bit vague as to what this looks like.

Beaton et al., (2000) have detailed a guideline on cross cultural adaptation of psychological scales which is also the method used in the Sri Lankan adaptation of the DAST-10. A similar approach can be applied for the Hindi version.

Figure 1

Graphic representation of the stages of cross-cultural adaptation recommended by Beaton et al.



This kind of an approach not only increases the quality of the translation but also ensures that the version is pretested before employing it for a larger sample. A more thorough adaptation process increases the likelihood of a strong correlation with the original version. This possibly

could save us time as the sample testing is more likely to be successful. Following pre-established adaptation guidelines such as this one increases the credibility of the study.

Methods (Measurement): Elaboration on the role of PHQ-9 and GAD-7 in the Study

It is unclear as to how the Patient Healthcare Questionnaire - 9 and General Anxiety Disorder - 7 scales help in validating the study. I think the idea is to determine how accurate the adapted scale is in measuring the underlying construct (substance abuse) as opposed to other comorbidities such as depression or anxiety.

Substance use disorders (SUD) and mental disorders commonly co-occur, with research indicating that about half of individuals with a mental illness will experience an SUD at some point, and vice versa. While data on youth comorbidity is less extensive, studies show that over 60% of adolescents in substance use treatment programs meet criteria for another mental illness. High comorbidity rates are observed between SUDs and various mental disorders, including anxiety disorders (generalized anxiety, panic disorder, and PTSD), depression, bipolar disorder, ADHD, psychotic disorders, borderline personality disorder, and antisocial personality disorder. Serious mental illnesses (SMI) like major depressive disorder which significantly impair life activities, also show a pronounced overlap with SUDs, with around 25% of individuals with SMI having an SUD. Approximately one in four people with SMI also have SUD (Bethesda, 2020).

These scales (PHQ-9 and GAD-7) enable us to classify participants based on whether they are undergoing comorbid conditions or SUD exclusively which in turn helps us assess the construct validity of our DAST-10. By adding this information to the research proposal, we can make it clear as to how exactly these scales help in this study.

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Week 10/11 Essay Task - Identifying Overlooked Research Gaps

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Limited Sample Population and Absence of Bilingual Population Testing

Widening Sample from Clinical Population to Non-Clinical Population

By focusing only on individuals already receiving treatment for Substance Use Disorder (SUD) at the Institute of Human Behavior and Allied Sciences (IHBAS), we risk creating a scale that may overly cater to those with more severe or easily detectable symptoms, potentially limiting its sensitivity and accuracy for lower risk individuals. This is particularly important for a screening tool like the DAST-10, as its utility lies in identifying a range of substance use severities, from low-level or preliminary symptoms to a clinically diagnosable SUD. A scale tested only on treatment-seeking individuals could fail to detect subtle symptoms in those at earlier stages of substance use, thus reducing its effectiveness as a broad spectrum screening tool.

Furthermore, it's crucial that the adapted items avoid any language that might come across as judgmental. However, individuals at IHBAS, who are already in treatment for substance use, are less likely to be influenced by social stigma around substance usage and are also less likely to find the need to be dishonest in answering to the scale (Henderson et al., 2013). In broader settings, though, people may downplay their substance use due to widespread stigma in India. Given our limited sample, it may be challenging to determine whether specific items on the scale trigger social desirability bias among participants by sounding stigmatized.

Therefore, Incorporating individuals from both clinical (treatment-seeking) and community (non-treatment-seeking) settings could potentially lead to better validation of the scale, refining it to differentiate more accurately across a wider range of substance use behaviors. This not only creates better generalizability but also helps avoid overpathologizing substance use

in populations that might not otherwise meet clinical SUD criteria, thereby ensuring the adapted DAST-10 can serve as an effective early intervention tool as well. A broader sample will also help us identify items that may provoke social desirability bias due to reasons mentioned in the previous paragraph.

Bilingual Testing for Cross-Language Validity

Testing the Hindi version of the DAST-10 alongside the original English version with a bilingual group ensures that the translated items retain their intended meaning and that the scale functions consistently across languages. Bilingual testing ensures that subtle differences in cultural or linguistic interpretation are identified and addressed (Lan et al., 2008). Moreover, if bilingual participants show high correlations in responses across both language versions, it provides empirical evidence of accurate translation, which can make the adapted tool more acceptable for clinical and research use across diverse linguistic groups within India.

Without bilingual testing, the adapted DAST-10 might face limitations in validity due to potential translation inaccuracies or cultural variables that were missed during adaptation. In such cases, the Hindi DAST-10 might not fully align with the original English version, potentially leading to variations in diagnostic outcomes that could undermine its reliability. Currently, we rely on the Alcohol, Smoking and Substance Involvement Screening Test (ASSIST) to validate our DAST-10 through comparison. Even if we infer overall accuracy of the scale through this method, we cannot precisely establish the equivalence of each item with the original scale.

Lack of Comprehensive Assessment and Diagnostic Methods

This limitation also highlights scope for further studies in this area. While the DAST-10 is an effective screening tool for substance use disorder (SUD) in various clinical and research settings, it is limited to screening rather than full assessment or diagnostic purposes. It doesn't cover all the DSM-5 criteria for SUD within its items (McCabe et al., 2006). The Indian healthcare setting, particularly in rural areas or lower-resource institutions, often lacks an integrated assessment system for mental health and SUD. Given this context, a sole reliance on DAST-10 screening may leave a gap in patient care and follow-up. Although DAST-10 provides preliminary indicators, it is not designed to diagnose or assess the severity or co-occurring conditions, which could lead to an incomplete understanding of patients' SUD profiles (Skinner, 1982).

By relying solely on DAST-10, we may not address the broader need for a diagnostic pathway that would help patients receive comprehensive care, particularly in areas where additional diagnostic tools are not readily available. Bridging the gap between screening and assessment could better patient outcomes by creating a more nuanced understanding of individual SUD cases, allowing for targeted intervention and treatment planning. Consequently, understanding the limitations of DAST-10 as a screening tool and recognizing the need for follow-up assessments could significantly improve the study's impact on patient care and mental health infrastructure.

It is also worth noting that even though we enable large amounts of the population to be screened effectively and efficiently, there are no further mechanisms for individuals suffering from SUD to actually avail the help they need, post screening. It then becomes imperative that

we push for better mental health facilities and infrastructure to eventually unlock practical benefits that we are currently unable to. Furthermore, when more people are employed under mental health care centers, who then enable more people to get treated, we also influence the society at large to be more accepting of these conditions. This helps gradually break social stigma by normalizing conversations around such issues. However, there are still a plethora of changes that we need to advocate for in India pertaining to the psychological wellbeing of individuals; ranging from adapting more assessment scales to improving the quality of treatment interventions in mental health institutions.

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Week 12 Essay Task - illegal activities in order to obtain substances

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Illegal Activities One Might Engage With In Order To Obtain Drugs

Burglary

Burglary involves breaking into homes, shops, or storage facilities to steal. In rural areas, houses and businesses often lack surveillance and security systems, making them easier targets. Individuals often target neighbors, or small businesses as larger financial corporations may be more risky to engage with. Agricultural equipment, food grains, and small amounts of cash may be stolen. Drugs can be expensive especially when you are perpetually buying them due to addiction; theft or burglary may make them affordable for these individuals.

Assault

Some individuals may resort to assaulting others for money or drugs directly. This can include violent robberies or intimidating others into giving up money or substances. In rural settings, where law enforcement may not be as stringent, individuals with SUD may use threats or physical force to exploit vulnerable members of their locality.

Drug Trafficking

Engaging in drug trafficking allows individuals to acquire drugs in exchange for distribution. In rural areas, they may serve as suppliers, transporting substances across remote regions where detection risks are lower. This is a mutually exploitative cycle where one furthers their own addiction from gaining access to drugs by being a dealer while they also target other vulnerable actors to buy the drugs.

Drug Manufacturing/Cultivation

Rural regions with high access to farmland or natural resources are particularly suited to cultivate certain drugs. Cultivation of *Ganja* (Cannabis), *Tobacco* and other such drugs are very common in India. The lack of regulatory oversight and law enforcement also make it easier to engage in such practices. This serves as a direct way to ensure a personal supply while also earning some income.

Fraud

Fraud includes activities like falsifying documents, forging signatures, or making false claims to access money, resources, or drugs. In rural areas, individuals may target local banks, cooperatives, or government schemes. They might also fake prescriptions to access medications with psychoactive properties. Fraud is likely because it offers a comparatively low-risk, high-reward way to sustain their habits.

Bribery

Bribery involves offering money, goods, or services to officials, shopkeepers, or other individuals to procure drugs or avoid legal consequences. In rural areas, underpaid law enforcement officers or corrupt officials might accept bribes to overlook illegal activities. This method of exploitation may be more common amongst higher status individuals.

Racketeering

Racketeering includes organizing or participating in a number of crimes to extract money or goods through illegal means, such as extortion or blackmail. In rural settings, individuals with SUD might coerce others into providing resources, threatening to expose secrets or harm their livelihoods. Such activities often emerge from desperation, as well as the absence of institutional support or alternative income opportunities.

Substance use impairs judgment and increases desperation, making all these crimes more likely as individuals prioritize acquiring drugs over maintaining social trust or morality.

Week 13 Essay Task - Flowcharts

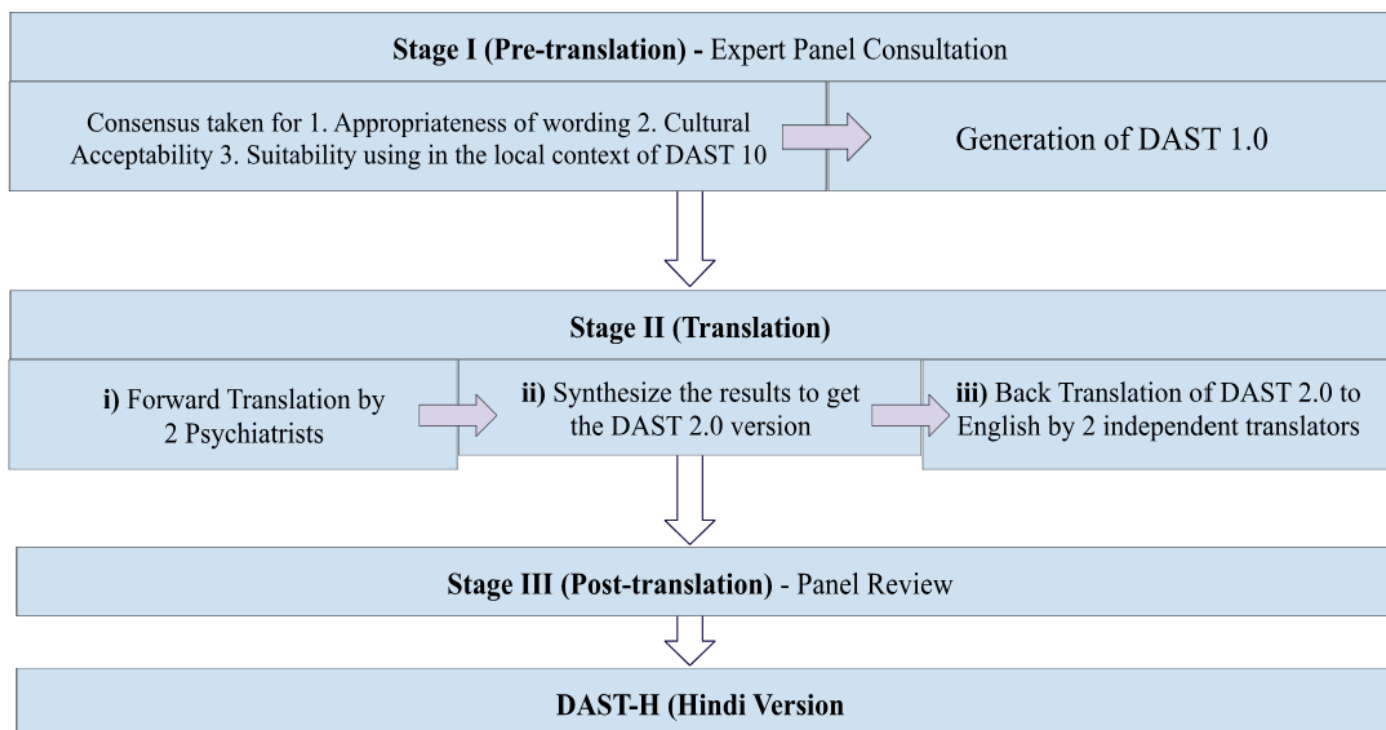
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Adaptation Flowchart



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